

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SwapSphere – A Smart Barter Platform with Intelligent Exchange Matching

1. Introduction

1.1 Purpose

SwapSphere is a web-based barter platform developed to simplify and modernize item exchange without the use of money. Unlike conventional online marketplaces that focus on buying and selling, SwapSphere enables users to exchange items directly with one another through an intelligent matching system.

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of the SwapSphere system. This document serves as a reference for the design, development, testing, and maintenance of the application. It also provides an overall understanding of the system's objectives, features, and operational requirements.

1.2 Scope

SwapSphere is developed using the MERN (MongoDB, Express.js, React.js, and Node.js) stack and is designed to support secure and efficient barter transactions.

The platform allows users to create item listings, specify preferred exchange items using prioritized categories and subcategories, browse available listings, and participate in direct or graph-based exchange suggestions. The system supports both two-way and three-way exchanges, provides communication through private Exchange Rooms, and maintains trust through ratings, reviews, and exchange history.

Guest users can browse listings without creating an account, while registered users can create listings, request exchanges, communicate with other users, and participate in completed exchanges.

1.3 Objectives

The main objectives of SwapSphere are:

- To provide a secure platform for exchanging items without monetary transactions.
- To simplify barter by automatically identifying possible exchange opportunities using graph-based matching.
- To support both direct exchanges and intelligent multi-user exchanges.
- To improve user trust through ratings, reviews, and exchange history.
- To provide an organized communication system for planning successful exchanges.

1.4 Definitions and Abbreviations

Term	Description
Barter	Exchange of goods without using money
MERN	MongoDB, Express.js, React.js, Node.js
Exchange Room	Private workspace created after an exchange request is accepted
Graph Matching	Algorithm used to identify possible exchange cycles
Guest User	User who can browse listings without logging in
Registered User	Authenticated user capable of creating listings and requesting exchanges
Administrator	User responsible for managing the platform

1.5 Technologies Used

The system is developed using the following technologies:

- React.js
- Node.js
- Express.js
- MongoDB
- Bootstrap
- Cloudinary
- Google OAuth
- Git and GitHub

1.6 Document Overview

This document describes the requirements and overall design of SwapSphere. It includes the system overview, functional requirements, non-functional requirements, external interfaces, system modules, constraints, assumptions, and overall project summary.

2. Overall Description

2.1 Product Perspective

SwapSphere is an independent web-based barter platform developed to facilitate item exchanges without the use of money. The application combines traditional marketplace browsing with an intelligent graph-based matching system to help users discover suitable exchange opportunities.

Unlike conventional online marketplaces that primarily focus on buying and selling, SwapSphere enables users to exchange items directly or participate in multi-user exchanges. The platform also provides communication and meeting coordination through private Exchange Rooms after an exchange request has been accepted.

The application follows a client-server architecture using the MERN stack. React.js is used to build the user interface, Node.js and Express.js handle the backend services, MongoDB stores application data, Cloudinary manages media uploads, and Google OAuth provides secure authentication.

2.2 Product Functions

The major functions provided by SwapSphere include:

- User registration and authentication using Email or Google Sign-In.
- Public browsing of marketplace listings.
- User profile management.
- Item listing and management.
- Category and subcategory based item organization.
- Priority-based exchange preference selection.
- Graph-based two-way and three-way exchange suggestions.
- Direct exchange requests through marketplace browsing.
- Exchange request management.
- Private Exchange Rooms for accepted exchanges.
- Real-time messaging and image sharing.
- Meeting scheduling between exchange participants.
- Notifications for important exchange activities.
- Ratings and reviews after successful exchanges.
- Administrative management of users, categories, and listings.

2.3 User Classes

The system consists of three types of users.

Guest User

Guest users can browse marketplace listings, search for items, apply category filters, and view public user profiles. They are not allowed to create listings, initiate exchange requests, or participate in exchanges.

Registered User

Registered users can create and manage item listings, specify exchange preferences, browse the marketplace, receive graph-based exchange suggestions, initiate direct exchanges, participate in Exchange Rooms, communicate with other users, schedule meetings, complete exchanges, and submit ratings and reviews.

Administrator

Administrators are responsible for maintaining the platform. They can manage users, categories, and listings, suspend or reactivate user accounts, and monitor platform activities. Administrators do not participate in exchanges unless they use a separate registered user account.

2.4 Operating Environment

SwapSphere is a web-based application that can be accessed using modern web browsers on desktop computers, laptops, tablets, and smartphones.

Client Side

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

Server Side

- Node.js Runtime
- Express.js
- MongoDB
- Cloudinary
- Google OAuth

2.5 Design Constraints

The development of SwapSphere is subject to the following constraints:

- The application is developed using the MERN stack.
- Internet connectivity is required for all major operations.
- Images are stored using Cloudinary.
- Only registered users can create listings and participate in exchanges.
- The current version supports only two-way and three-way exchanges.
- Administrative functions are restricted to authorized administrators.

2.6 Assumptions and Dependencies

The successful operation of SwapSphere depends on the following assumptions:

- Users have access to a stable internet connection.
- Users provide accurate information while creating profiles and listings.
- Google OAuth services remain available for authentication.
- Cloudinary remains available for media storage.
- MongoDB is available for storing application data.
- Users cooperate honestly during exchanges and provide fair ratings after successful transactions.

3. Functional Requirements

3.1 User Registration and Authentication

The system shall provide secure authentication to ensure that only authorized users can access exchange-related features.

The system shall:

- Allow users to register using email and password.
- Allow users to sign in using Google OAuth.
- Ensure that each email address is unique.
- Allow authenticated users to log in and log out securely.
- Maintain user sessions after successful authentication.
- Allow guest users to browse listings without logging in.
- Restrict exchange-related features to registered users only.

- Restrict administrative functions to authorized administrators.

3.2 User Profile Management

Each registered user shall have a personal profile that displays account information and trust indicators.

The system shall:

- Allow users to view and edit their profiles.
- Allow users to upload or update a profile picture.
- Display public user profiles.
- Display trust indicators including:
 - Total completed exchanges
 - Average rating
 - Exchange success rate
 - Public exchange history
 - Reviews received
- Automatically update trust indicators after completed exchanges.

3.3 Marketplace Browsing

The marketplace allows users to discover available items and manually initiate exchanges.

The system shall:

- Display all available item listings.
- Support keyword-based search.
- Support category and subcategory filtering.
- Allow users to sort listings.
- Display detailed item information.
- Display the owner's public profile and trust indicators.
- Allow guest users to browse listings only.
- Allow registered users to initiate direct exchange requests by offering one of their own active listings.

3.4 Item Listing Management

Registered users can create and manage their own listings.

The system shall:

- Allow users to create new item listings.
- Allow users to edit or remove their listings.
- Allow multiple image uploads.
- Allow optional video uploads.
- Allow users to select a category and an optional subcategory.
- Allow users to specify multiple exchange preferences arranged by priority.
- Maintain the item status throughout the exchange lifecycle.
- Consider only AVAILABLE items for graph matching.

3.5 Graph-Based Exchange Matching

The graph matching engine analyzes all available listings to identify potential barter opportunities.

The system shall:

- Analyze all AVAILABLE listings.
- Compare exchange preferences using categories and optional subcategories.
- Identify valid two-way exchange cycles.
- Identify valid three-way exchange cycles.
- Display direct and three-way exchange suggestions separately.
- Rank suggestions according to exchange priority.
- Calculate a combined exchange score for three-way exchanges.
- Notify users when new exchange suggestions become available.
- Generate suggestions only; exchange requests are created only after user confirmation.

3.6 Exchange Request Management

Exchange requests represent a user's intention to exchange one or more items.

The system shall:

- Allow users to create exchange requests from graph suggestions.
- Allow users to create direct exchange requests from marketplace listings.

- Notify all participants about new exchange requests.
- Allow participants to accept or reject requests.
- Create an Exchange Room only after all participants accept.
- Allow the initiator to cancel pending requests.
- Expire requests that remain unanswered beyond the specified time limit.
- Update exchange success rates appropriately after cancellations.

3.7 Exchange Room Management

Exchange Rooms provide a private environment for participants to coordinate exchanges.

The system shall:

- Automatically create an Exchange Room after all participants accept.
- Restrict access to participating users only.
- Display participating users and exchanged items.
- Allow meeting scheduling.
- Store meeting information.
- Allow participants to confirm exchange completion.
- Mark items as EXCHANGED only after every participant confirms.

3.8 Messaging

The messaging module enables communication between exchange participants.

The system shall:

- Allow text messaging.
- Allow image sharing.
- Restrict messages to Exchange Room participants.
- Display messages in chronological order.
- Record sender information and timestamps.
- Generate system messages for important events.

3.9 Notifications

The notification system keeps users informed about important platform activities.

The system shall:

- Notify users about new exchange requests.
- Notify users about accepted, rejected, cancelled, or expired requests.
- Notify users about new graph-based exchange suggestions.
- Notify participants about meeting updates.
- Notify users after successful exchanges.
- Allow users to view and manage notifications.

3.10 Ratings and Reviews

The rating system helps establish trust among users.

The system shall:

- Allow ratings only after completed exchanges.
- Allow users to rate every participant involved in the exchange.
- Support ratings on a five-star scale.
- Allow optional written reviews.
- Calculate average ratings automatically.
- Display reviews and trust indicators on public profiles.
- Restrict reviews to users who participated in the same completed exchange.

3.11 Administration

The administration module is responsible for maintaining the platform.

The system shall:

- Provide secure administrator login.
- Allow administrators to manage users.
- Allow administrators to suspend or reactivate user accounts.
- Allow administrators to manage categories.
- Allow administrators to moderate item listings.
- Allow administrators to monitor exchange requests and Exchange Rooms.

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the SwapSphere system.

4.1 Performance Requirements

- The system shall respond to user requests within an acceptable time under normal operating conditions.
- Item searches and marketplace browsing shall provide results without noticeable delay.
- Graph-based exchange suggestions shall be generated efficiently for available listings.
- The system shall support multiple users accessing the platform simultaneously.

4.2 Security Requirements

- User authentication shall be required for all exchange-related activities.
- Passwords shall be securely encrypted before being stored.
- Sensitive communication between the client and server shall be performed over HTTPS.
- Users shall only be permitted to access resources for which they are authorized.
- Exchange Rooms shall be accessible only to participating users.
- Administrative functions shall be restricted to administrator accounts.

4.3 Reliability Requirements

- The system shall maintain data consistency throughout the exchange process.
- Exchange requests and user information shall be stored reliably.
- The system shall recover gracefully from unexpected failures whenever possible.
- Data loss during normal system operation shall be minimized.

4.4 Availability Requirements

- The application should be available whenever internet connectivity and server services are operational.
- Users should be able to access the platform using modern web browsers without requiring additional software.

4.5 Usability Requirements

- The interface shall be simple, responsive, and easy to navigate.

- Users shall be able to perform common tasks with minimal training.
- The application shall provide clear feedback for user actions.
- The interface shall remain consistent across different pages.

4.6 Scalability Requirements

- The system architecture shall support future expansion.
- The graph matching module shall be designed to support future N-way exchanges.
- Additional categories, subcategories, and features shall be incorporated without major architectural changes.

4.7 Maintainability Requirements

- The application shall follow a modular architecture to simplify maintenance.
- Source code shall be organized to support future enhancements.
- Individual modules should be updatable without affecting unrelated components.

4.8 Compatibility Requirements

The system shall support modern web browsers including:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

The application shall function on desktop computers, laptops, tablets, and smartphones.

4.9 Portability Requirements

- Since SwapSphere is a web application, users shall be able to access it from any supported device with an internet connection.
- No platform-specific software installation shall be required.

5. External Interface Requirements

5.1 User Interface

SwapSphere provides a responsive and user-friendly web interface designed to support both desktop and mobile devices. The interface is developed using React.js and Bootstrap to ensure consistency and ease of use across different screen sizes.

The major user interface components include:

- Home page displaying featured and recent item listings.
- Marketplace page with search, filtering, and category browsing.
- User registration and login pages.
- User profile page displaying profile information, trust indicators, and reviews.
- Item listing page for creating and managing listings.
- Item details page showing item information, exchange preferences, and owner details.
- Graph-based exchange suggestions page.
- Direct exchange request interface.
- Exchange Room with messaging and meeting planning.
- Notifications page.
- Administrator dashboard.

The interface is designed to be simple, intuitive, and easy to navigate for all users.

5.2 Hardware Interface

SwapSphere does not require any specialized hardware.

Client Device

The application can be accessed using:

- Desktop computers
- Laptops
- Tablets
- Smartphones

Users require a keyboard, mouse, or touchscreen along with an active internet connection.

Server

The server should support:

- Node.js Runtime
- MongoDB Database
- Internet connectivity for communication with Cloudinary and Google OAuth services

5.3 Software Interface

SwapSphere interacts with the following software components and third-party services.

Software	Purpose
React.js	Frontend development
Bootstrap	Responsive user interface
Node.js	Backend runtime
Express.js	REST API development
MongoDB	Database management
Cloudinary	Image and media storage
Google OAuth	User authentication
Git & GitHub	Version control

5.4 Communication Interface

Communication between the client and server is performed using RESTful APIs over the HTTP/HTTPS protocol.

The system communicates with external services for:

- User authentication through Google OAuth.
- Image upload and retrieval using Cloudinary.
- Data storage and retrieval using MongoDB.
- Communication between the React frontend and Express backend.

All sensitive communication shall be performed using secure HTTPS connections.

6. System Modules

SwapSphere is divided into several functional modules, each responsible for a specific part of the system. The modular design improves maintainability, scalability, and ease of future development.

6.1 Authentication Module

This module manages user registration, login, logout, and authentication using email/password and Google OAuth. It ensures that only authenticated users can access protected features of the platform.

6.2 User Management Module

This module allows users to manage their personal profiles, update profile information, upload profile pictures, and view trust indicators such as ratings, completed exchanges, and exchange history.

6.3 Marketplace Module

The marketplace displays all available item listings and allows users to search, filter, and browse items based on categories and subcategories. Registered users can initiate direct exchange requests from this module.

6.4 Item Management Module

This module enables registered users to create, edit, and remove item listings. Each listing includes item details, media, category, optional subcategory, and prioritized exchange preferences.

6.5 Graph-Based Exchange Matching Module

This module analyzes all available listings and identifies possible barter opportunities using a graph-based matching algorithm.

The module:

- Identifies direct (two-way) exchanges.
- Identifies three-way exchange cycles.
- Ranks exchange suggestions based on user priorities.
- Displays direct and three-way suggestions separately.

6.6 Exchange Management Module

This module handles the complete exchange request process. It manages request creation, participant responses, acceptance, rejection, cancellation, and the transition to an Exchange Room after successful acceptance.

6.7 Exchange Room Module

After an exchange request is accepted by all participants, an Exchange Room is created. This module provides a private environment where participants can communicate, schedule meetings, and confirm exchange completion.

6.8 Messaging Module

This module enables participants to exchange text messages and images within an Exchange Room. It also generates system messages for important events such as meeting creation and exchange completion.

6.9 Notification Module

The notification module informs users about important activities, including new exchange requests, exchange suggestions, meeting updates, cancellations, and completed exchanges.

6.10 Rating and Review Module

After a successful exchange, participants can rate and review one another. This module calculates trust indicators such as average rating and exchange success rate, which are displayed on public user profiles.

6.11 Administration Module

The administration module allows authorized administrators to manage users, categories, and item listings. Administrators can suspend user accounts, moderate inappropriate listings, and maintain the overall integrity of the platform.

7. Assumptions and Constraints

7.1 Assumptions

The successful operation of SwapSphere depends on the following assumptions:

- Users have access to a stable internet connection.
- Users provide accurate information while creating profiles and item listings.
- Google OAuth services remain available for user authentication.
- Cloudinary services remain available for storing uploaded media.
- MongoDB is available for storing and retrieving application data.
- Users cooperate honestly during exchanges and provide fair ratings after successful transactions.
- Exchange participants complete exchanges only after mutual agreement.

7.2 Constraints

The development of SwapSphere is subject to the following constraints:

- The application is developed using the MERN (MongoDB, Express.js, React.js, Node.js) stack.
- The current version supports only two-way and three-way barter exchanges.
- Internet connectivity is required for all major system functionalities.
- Images and videos are stored using Cloudinary.
- Only registered users are permitted to create listings, initiate exchange requests, and participate in exchanges.
- Administrative functions are restricted to authorized administrators.
- The graph matching algorithm considers only items with **AVAILABLE** status while generating exchange suggestions.

8. Conclusion

SwapSphere is designed to provide a secure and intelligent platform for exchanging items without monetary transactions. By combining traditional marketplace browsing with graph-based exchange matching, the system simplifies the process of finding suitable exchange partners while improving transparency and user trust.

The platform supports direct and multi-user barter exchanges, secure communication, meeting coordination, notifications, and a reputation system based on completed exchanges and user reviews. Its modular architecture allows future enhancements, including support for larger exchange cycles and additional platform features.

The requirements presented in this document provide a clear foundation for the design, implementation, testing, and future maintenance of the SwapSphere application.